Candidate seat no: ________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Second semester of B.Pharm. Examination April - May 2018
MA122 Advanced Mathematics

Date: 14.05.2018, Monday **Time:10:00a.m. to 01:00p.m.** **Maximum Marks: 80**

Instructions:

1. **There are three sections in this question paper.**
2. **SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.**
3. **Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.**
4. **Figures to right indicate marks.**
5. **Draw neat labeled sketches wherever necessary.**
6. **Use of scientific calculator is allowed.**

SECTION - I

Q 1 Choose the correct answer for the following questions. **20**

1. Fourth derivative of $y = e^{5x}$ is

[A] $y_4 = 2^4 e^{5x}$.

[B] $y_4 = 4^4 e^{5x}$.

[C] $y_4 = e^{3x}$.

[D] $y_4 = 5^4 e^{5x}$.

2. Second derivative of $y = x + 2$ is

[A] 0.
[B] 1.
[C] 2.
[D] 2.

3. Which of the following is true?

[A] Every Taylor series is Macluarin series.
[B] Macluarin series expansion around 0 is Taylor series.
[C] Macluarin series expansion arround 1 is Taylor series.
[D] Macluarin series expansion arround 2 is Taylor series.

4. The order and degree of an ordinary differential equation $y'' + y - 3 = 0$ are

[A] order 1 and degree 1.
[B] order 2 and degree 1.
[C] order 1 and degree 2.
[D] order 2 and degree 2.

5. The first order ordinary differential equation $(2x+ay)\,dx+(3x+4y)\,dy=0$ is exact if and only if a is

[A] 1.

[B] 2.

[C] 3.

[D] 4.

6. The general form of linear differential equation is

[A] $\frac{dy}{dx}+p(y)y=q(x,y)$.

[B] $\frac{dy}{dx}+p(x,y)y=q(y)$.

[C] $\frac{dy}{dx}+p(x)y^n=q(x),\ n>1$.

[D] $\frac{dy}{dx}+p(x)y=q(x)$.

7. Order of homogeneous function $f(x,\ y)=x^3+y^3$ is

[A] 2.

[B] 3.

[C] 4.

[D] 5.

8. Which of the following is a root of auxiliary equation of $y''+2y'+y=0$?

[A] -1

[B] 0

[C] 2

[D] 4

9. If roots of auxiliary equation are 1 and 2, then corresponding component (basis) solutions of a 2nd order ordinary differential equation are

[A] e^x & e^{2x}.

[B] e^{-x} & e^x.

[C] $\cos x$ & $\sin x$.

[D] $e^x\cos x$ & $e^x\sin x$.

10. Which of the following is true for the rate of change in number of bacteria(x) with respect to time(t) is inverse propositional to t^2?

[A] $t\frac{dx}{dt}=m$

[B] $t^2\frac{dx}{dt}=m$

[C] $\frac{dx}{dt}=mt^2$

[D] $\frac{dx}{dt}=mt+1$

11. The rate at which bacteria multiply is proportional to the current number of bacteria. if x be the number of bacteria present at time t, then

[A] $\frac{dx}{dt} = mx^2$.

[B] $\frac{dx}{dt} = mx$.

[C] $\frac{dx}{dt} = mx + kt$.

[D] $\frac{dx}{dt} = e^{mx}$.

12. If T be the temperature of the body at time t and t_s is the temperature of surrounding, then Newton's law of cooling is given by

[A] $\frac{dT}{dt} = k(T - t_s)$.

[B] $\frac{dT}{dt} = kT + t_s$.

[C] $\frac{dT}{dt} = kT$.

[D] $\frac{dT}{dt} = k(t_s - T)$.

13. Equation of a straight line passing through (1, 0) and (0, 1) is

[A] $x - y = 2$.

[B] $x + y = 1$.

[C] $3x + y = 2$.

[D] $x - 3y = 5$.

14. Equation of a straight line parallel to $x + 2y = 3$ is

[A] $x + 2y = 4$.

[B] $y = 5x + 8$.

[C] $y = 2x + 3$.

[D] $y = 4x + 5$.

15. Two lines with slope $m_1 \& m_2$ are said to be perpendicular if

[A] $m_1 m_2 = 1$.

[B] $m_1 - m_2 = 0$.

[C] $m_1 + m_2 = 1$.

[D] $m_1 m_2 = -1$.

16. Distance between two points (1,0) and (2, 0) is

[A] 0.

[B] 1.

[C] 2.

[D] 3.

17. Mean of data set 2, 6, 7, 7, 8, 12, 14 is

[A] 7.

[B] 8.

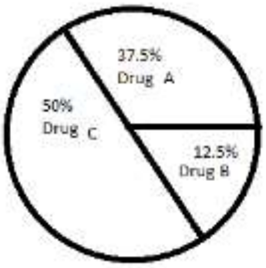
[C] 9.

[D] 10.

18. Data set 1, 2, 3, *a*, 15, 16 and 17 is arranged in increasing order. If Median of a following data set is 9, then *a* is

[A] 8.
[B] 9.
[C] 10.
[D] 11.

19. Following graph shows production of different types of drug by a company in a month. If total production of drugs by a company is 50 kg in one month, then production of drug C in a month is



[A] 20 kg.
[B] 25 kg.
[C] 30 kg.
[D] 35 kg.

20. If mean of data set 2, 5, 8, 9, 10, a, 15 and 19 is 10 , then value of *a* is

[A] 10.
[B] 12.
[C] 14.
[D] 15.

SECTION – II

Q 2 Attempt any **TWO** of the following

A If rate of a change of number of Bacteria(x) with restpect to t-time(hour) is given by $\frac{dx}{dt} = xt$, then find number of Bacteria at t = 3 hours and at intial stage $t = 1$ *and* $x = 1$. **05**

B Find mean, median and relative frequency distribution(%) of ungrouped data set 1, 4, 5, 3, 6, 7, 8, 2 and 2. **05**

C Derive equation of the distance between two points and using it find distance between (0, 1) and (5, 4). **05**

Q 3 Attempt any **TWO** of the following

A Verify Cauchy's mean value theorem for $f(x) = \log x$ and $g(x) = \frac{1}{x}$, $x \in [1, 2]$. **05**

B	Identify the type of an ordinary differential equation $\frac{dy}{dx}+y=x$ and find particular solution with help of intial condition $y(0)=1$.	**05**
C	If rate of production of drug(y) in kg with resptet to time(x) in hour by machine is given by a differential equation $y''+y=0$. Find *y* at any time *x*.	**05**

SECTION – III

Q 4	Attempt any **FOUR** of the following	
A	Solve $y''+4y'+4y=0$.	**05**
B	If the temperature of a body drops from 80° to 70° in one minute when the temperature of the surrounding is 30°,what will be the temperature of the body at the end of the second minute?	**05**
C	Find Mean for following ungrouped data 2, 6, 11, 9, 19, 13, 20, 21, 24, 31, 32, 29 and 42. Also organize the above data into groups with 5 intervals starting from smallest value and using it evaluate relative frequency(%) of each class.	**05**
D	A rectangle having vertices A(0, 0), B(0, 3), C(2, 0) and D(2, 3) is given. Find length, height and area of rectangle.	**05**
E	Solve $(x^2+y^2)dx+2xydy=0$.	**05**
F	**(I)** State the governing differential equation of Newton's law of cooling and solve it.	**03**
	(II) Describe mean and mode.	**02**
Q 5	Attempt any**FOUR**of the following	
A	State Taylor's theorem for expansion of a function $f(x)$ at $x=a$ and using it expand $f(x)=\cos x$ at $x=0$.	**05**
B	State Maclaurin's Series and using it find Maclaurin's Series of $f(x)=e^x$.	**05**
C	Solve : $\frac{dy}{dx}+2\frac{y}{x}=x$	**05**
D	Show that $2xydx+(x^2+y^2+3)dy=0$ is an exact differential equation and obtain general solution of it.	**05**
E	Find particular integral of a differential equation having basis solutions $y_1=e^x$, $y_2=e^{2x}$ and $R(x)=e^{-x}$.	**05**
F	Solve: $y''+5y'+6y=0$.	**05**

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